Prolog Programming for Artificial Intelligence
4th Edition
—
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1 Introduction to Prolog

1.1 Defining relations by facts

1.1

(a)
```
?- parent( jim, X).
false.
```

(b)
```
?- parent( X, jim).
X = pat.
```

(c)
```
?- parent( pam, X), parent( X, pat).
X = bob.
```

(d)
```
?- parent( pam, X), parent( X, Y), parent( Y, jim).
X = bob,
Y = pat.
```

1.2

(a)
```
?- parent( X, pat).
X = bob.
```

(b)
```
?- parent( liz, X).
false.
```

(c)
```
?- parent( X, pat), parent( Y, X).
X = bob,
Y = pam ;
X = bob,
Y = tom.
```

1.2 Defining relations by rules

1.3

(a)
```
happy( X) :- hasachild( X).
```

(b)
```
hastwochildren( X) :-
        hasachild( X),
        parent( X, Y),
        sister( Y, Z).
```

1.4

```
grandchild( X, Z) :-
    parent( Y, X),
    parent( Z, Y).
```

1.5

```
aunt( X, Z) :-
    parent( Y, Z),
    sister( Y, X).
```

1.3 Recursive rules

1.6

Yes, this is a correct definition of `ancestor`. In Figure 1, we have illustrated this alternative definition.

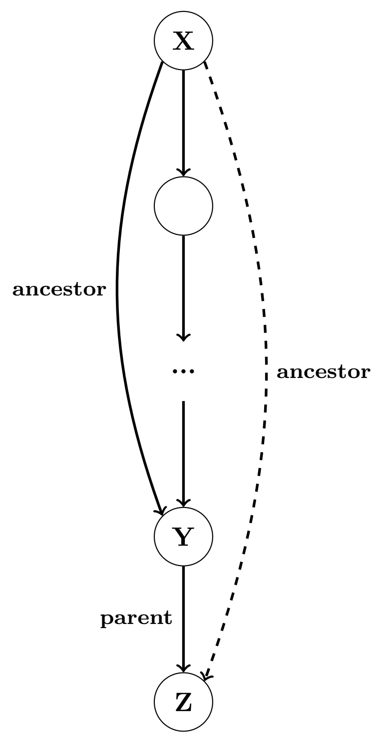

Figure 1: An alternative recursive definition of the `ancestor` relation.